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BIOS 702
Exercise 3.2
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One Sample Inference for a Binary Outcome

Question 1

The clinical trial by Jabbour compares a first-generation tyrosine kinase inhibitor (imatinib) to a third-generation, ponatinib. In the discussion of the manuscript, it is stated that "While it is unclear how the current findings compare with second-generation tyrosine kinase inhibitors in the absence of a head-to-head randomized trial, the MRD negativity rate at the end of induction in the imatinib group in PhALLCON (22%) was comparable with previous reports for second-generation tyrosine kinase inhibitors (14%-18%). We want to run a one-sided, one-sample proportion test to test if the MRD negativity rate at the end of induction for ponatinib is greater than previous reports for second-generation tyrosine kinase inhibitors (let's use the midpoint of 16%).

1. First, explain why it is acceptable in this situation for the test to be 1-sided.

   A one-sided test is appropriate because we are specifically interested in whether the MRD negativity rate for ponatinib is higher than historical rate of 16%, rather than whether it is simply different from 16%

2. From the manuscript, what is the MRD negativity rate at the end of induction for ponatinib from the study? (Hint: Check Table 2)

   The MRD negativity rate at the end of induction was 43% (61 of 142 patients) in the ponatinib group.

3. Run a one-sided one-sample test for the proportion, using the null hypothesis that $\pi \leq 16\%$. Do not use the continuity correction.

```
prop.test(61, 142, p = 0.16,
          alternative = "greater",
          correct = FALSE)
```

```
        1-sample proportions test without continuity
        correction

data:  61 out of 142, null probability 0.16
X-squared = 76.781, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is greater than 0.16
95 percent confidence interval:
 0.3631946 1.0000000
sample estimates:
        p
0.4295775

>
```

4. Write a definition of the 1-sided p-value in the context of this problem.

   If the true MRD negativity rate were 16%, the one-sided p-value is the approximate probability of obtaining a sample rate as high as or higher than the observed 61/142 (43%) in a sample of 142 patients.

Question 2

1. Use the normal approximation to verify “by hand” that the function above gives the same test statistic and p-value.

```
> phat = 61/142
> p0 = 0.16
> n = 142
>
> Z =  (phat - p0) / sqrt(p0*(1-p0)/n)
> Z^2
[1] 76.78144
>
> pnorm(Z, lower.tail = FALSE)
[1] 9.548199e-19
>
```

2. Interpret the confidence interval you see from the output of the one-sided hypothesis test. If you were to run a two-sided test, what would you have to set the conf.level at to see the same lower bound? Run that test to check.

```
prop.test(61, 142, p = 0.16,
          alternative = "two.sided",
          conf.level = 0.90,
          correct = FALSE)
```

```
        1-sample proportions test without continuity
        correction

data:  61 out of 142, null probability 0.16
X-squared = 76.781, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.16
90 percent confidence interval:
 0.3631946 0.4985937
sample estimates:
        p
0.4295775

>
```

   The 95% one-sided confidence interval has a lower bound of 0.3632, suggesting that the true MRD negativity rate for ponatinib is above approximately 36.3%. The 90% two-sided confidence interval has the same lower bound of 0.3632

Question 3

The study now includes 3 groups; first-, second-, and third-generation tyrosine kinase inhibitors. The investigators have the raw data from the first-generation drug, but we must rely on the literature to obtain summary results for the second-generation drug. What are the advantages and disadvantages of these two ways to make comparison groups? What biases might result?

The imatinib group is a stronger comparison group because it was randomized within the same trial as the ponatinib group. The investigators also have patient-level data, so they can examine differences in patient characteristics and use a consistent outcome definition. However, restricting the MRD analysis to patients with evaluable samples could introduce selection bias if missing samples are related to treatment or response.

Published results for second-generation inhibitors offer a useful historical benchmark without conducting another trial. Their main limitation is that only summary results are available: the investigators cannot adjust for patient-level differences or ensure that eligibility criteria, accompanying treatments, and MRD measurements were the same across studies. These differences could confound the comparison. Publication or selective reporting of favorable results could also bias the historical estimate. Because the second-generation group was not randomized against ponatinib, a difference in observed rates cannot be attributed to the drugs alone.

Question 4

Now, we want to compare the observed MRD negativity rate by end of induction for imatinib with the historical summary estimate of 16% for second-generation tyrosine kinase inhibitors. Specifically, we are interested in determining if imatinib was a good control and exhibited the MRD negativity rate expected of earlier generation drugs.

1. Would it make more sense to perform this hypothesis test as a one-sided or two-sided test and why?

   A two-sided test is appropriate because we want to assess whether the imatinib MRD negativity rate differs from the historical rate of 16% in either direction. A rate substantially higher or lower than 16% would both call into question whether imatinib showed the expected response rate.

2. Run this test. Interpret the results.

```
prop.test(15, 68, p = 0.16,
          alternative = "two.sided",
          correct = FALSE)
```

```
        1-sample proportions test without continuity
        correction

data:  15 out of 68, null probability 0.16
X-squared = 1.8573, df = 1, p-value = 0.1729
alternative hypothesis: true p is not equal to 0.16
95 percent confidence interval:
 0.1384901 0.3325674
sample estimates:
        p
0.2205882
```

The observed MRD negativity rate for imatinib was 15/68 (22.1%). The two-sided test gave χ^2 = 1.857 and p = 0.1729. Since p > 0.05, there is insufficient evidence that the imatinib rate differs from the historical rate of 16%. This result is consistent with imatinib showing a rate in the expected range, but it does not prove that the two rates are equal or that imatinib is a suitable control.

Question 5

Finally, let's say we are interested in the testing the hypothesis that the proportion of loss-of-response events among those with MRD negativity before induction in the imatinib group is equal to 0.25 (Hint: Check Table 2 in the paper).

1. What is the total sample size for this test and what is the sample proportion?

   The sample size is 13, and the sample proportion of loss-of-response event is 6/13 = 0.462 (46.2%)

2. The normal approximation test is not useful in this case because we have such a small sample. So, you will use the exact binomial test instead. Show the probability distribution for 0 to 13 loss-of-response events under H0. Verify that the distribution sums to 1. Hint: you can use the dbinom function.

```
events <- 0:13
probs <- dbinom(events, size = 13, prob = 0.25)

data.frame(events, probs)
sum(probs)
```

```
> data.frame(events, probs)
   events        probs
1       0 2.375726e-02
2       1 1.029481e-01
3       2 2.058963e-01
4       3 2.516510e-01
5       4 2.097092e-01
6       5 1.258255e-01
7       6 5.592245e-02
8       7 1.864082e-02
9       8 4.660204e-03
10      9 8.630008e-04
11     10 1.150668e-04
12     11 1.046062e-05
13     12 5.811453e-07
14     13 1.490116e-08
> sum(probs)
[1] 1
>
```

3. Find the two-sided p-value using the null hypothesis distribution you printed out in the question above.

```
sum(probs[c(1, 7:14)])
```

```
> sum(probs[c(1, 7:14)])
[1] 0.1039699
>
```

The two-sided exact p-value is P(X=0) + P(X ≥ 6) = 0.1040 under the null distribution.

4. Now find the p-value using the binom.test function. Both methods should give you the same answer.

```
> binom.test(6, 13, p = 0.25, alternative = "two.sided")

        Exact binomial test

data:  6 and 13
number of successes = 6, number of trials = 13,
p-value = 0.104
alternative hypothesis: true probability of success is not equal
to 0.25
95 percent confidence interval:
 0.1922324 0.7486545
sample estimates:
probability of success
             0.4615385
```

The binom.test() p-value is approximately 0.1040, matching the p-value calculated from the null distribution in Question 5.3.